MUNER AE PROJECT

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DOCUMENT VERSIONS

Version	Description
1.0	Starting release
1.1	Added rotary switch request

THE PROJECT WORK

The activity includes:

- Form a team and assign roles within the team
- Design the given system
- The initial project specifications are not final, they must be completed by analyzing the project's purpose and discussing them directly with the customer.
- The deliverables include at least the following:
    - Project details: block diagram, architectural choices, schematics, PCB, simulations, BOM, code overview, a working prototype, datasheet, other
    - Project management details: role assignment, budget management, planning, collaboration tools if used, other

DESCRIPTION OF THE PROJECT

Design of a steering wheel electronic board

The project consists of designing an electronic board for use as a steering wheel. The board must be able to read four electromechanical buttons (for UP, DW, and two other spare functions), read a communication line from which two temperatures are received, and read two analog inputs on the board itself.
The board must have a display to show messages when the buttons are pressed. In addition, it should show the temperatures received via the communication line and the value of the analog input it reads.
The boards should have nr. 2 LEDs (they can be driven via CAN bus).

SPECIFICATIONS

Parameter	Min	Typ	Max	Units	Notes
Power Supply					
Input Voltage (V_in)	5	12	16	V	
Standby Current (I_standby)	-	10	50	mA	When not active - display off
Operating Current (I_op)	-	100	<500	mA	During normal operation
Digital Inputs (Buttons)					
Number of Inputs	-	4	-	-	UP & DW gearshift, and nr.2 spare functions
Input Logic High Voltage (V_IH)				V	
Input Logic Low Voltage (V_IL)				V	
Analogue Inputs					
Number of Inputs		2			Nr.1 for clutch position target Nr.1 rotary switch zt N discrete positions for parameters setting
Input Voltage Range				V	
Communication Line					
Protocol	-	-	-	-	CAN or CANFD
Data Rate	-	-	-	bps/Hz	1 Mbps if standard, 4 Mbps if FD
Display					
Type	-	-	-	-	
Resolution	-	128x64	-	pixels	Suggested, but even higher resolution is appreciated
Backlight Current	-	-	-	mA	
LED					
Number of LEDs		2			Alarms or signals
Environmental					
Operating Temperature	-20	-	85	°C	
Storage Temperature	-40	-	125	°C	

NOTES

Where possible:

- Use of qualified automotive components
- Use of protections and robust design procedures
- Describe the testing, calibration procedure
- Describe the validation process
- Provide graphic interface on a PC to interact with the board

The use of cooling fans is not allowed
The prototype must be built with the official BOM parts when available.